

When structure makes the inverse cheap

Each of these is invertible under a condition readable off the matrix, and each has an inverse that inherits the same structure. Recognising the type is worth more than any general algorithm.

THE INVERSE IS FREE

1



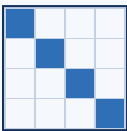
Orthogonal | $Q^T Q = I$

INVERTIBLE WHEN	INVERSE	COST
always — $\det Q = \pm 1$	Q^T	none — transpose

The only row where no arithmetic happens at all. Always invertible, and the inverse is a relabelling of the entries — which is why **orthogonal factors** are worth arranging for in a factorization.

THE INVERSE IS ENTRYWISE OR BLOCKWISE

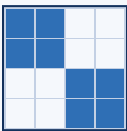
2



Diagonal | $D = \text{diag}(d_1, \dots, d_n)$

INVERTIBLE WHEN	INVERSE	COST
every $d_i \neq 0$	$\text{diag}(1/d_1, \dots, 1/d_n)$	n divisions

Reciprocate the diagonal. The condition is visible without computing anything, since the determinant is the product of exactly those entries.



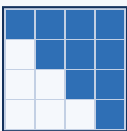
Block diagonal | $\text{diag}(A_1, \dots, A_k)$

INVERTIBLE WHEN	INVERSE	COST
every block A_i invertible	$\text{diag}(A_1^{-1}, \dots, A_k^{-1})$	invert each block

The diagonal case with numbers replaced by blocks. An $n \times n$ matrix in k equal blocks costs k inversions of size n/k rather than one of size n — a saving of k , since the cost is cubic.

THE INVERSE COMES FROM SUBSTITUTION

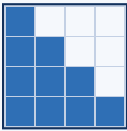
2



Upper triangular | zeros below the diagonal

INVERTIBLE WHEN	INVERSE	COST
every diagonal entry $\neq 0$	upper triangular	back substitution, n

Solve for the columns from the bottom up. The inverse stays upper triangular, which is why an **LU factorization** is worth keeping rather than discarding after the first solve.



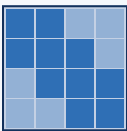
Lower triangular | zeros above the diagonal

INVERTIBLE WHEN	INVERSE	COST
every diagonal entry $\neq 0$	lower triangular	forward substitution, n

The mirror image, worked from the top down. Together with the row above, these are the two solves an LU-based inverse actually performs.

STRUCTURE PRESERVED BUT NO SHORTCUT

1



Symmetric | $A = A^T$

INVERTIBLE WHEN	INVERSE	COST
$\det(A) \neq 0$	also symmetric	half of a general solve

The one row whose condition is not local — symmetry constrains the inverse without deciding whether it exists. The saving is real but modest: only one triangle need be computed. When the matrix is also positive definite, **Cholesky** halves it again.

Two things recur down the table. The invertibility condition is always local — a diagonal entry, a block, never a determinant — and the inverse always belongs to the same family, which is what makes these routes composable. General **row reduction** discards both advantages and should be the fallback rather than the first move.